Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in analytical characterization, LC-MS/GPC/SEC-adjacent workflows, stability analysis, method documentation, and DOE-guided process development. Builds reproducible analytical data packages and SOP-ready methods that connect complex characterization results to product understanding and troubleshooting decisions.

CORE COMPETENCIES

LC / LC-MS Method Bridge | Analytical Development | RNA / LNP Characterization Bridge | Impurity & Stability Analysis | Method Lifecycle Documentation | CMC Troubleshooting | DOE / SPC | Technical Reports

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided analytical readout design for bio-based polymer process development, defining reproducibility criteria, process windows, and SOP-ready specifications.
- Translate FT-IR, thermal-analysis, spectroscopy, and process data into technical recommendations and repeatable methods for product/process understanding.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed, purified, and characterized 15+ organic material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC/SEC, TGA, DSC, and MS-informed analysis.
- Built structured datasets and written reports connecting chemical structure, purity, stability, and analytical signals to performance and troubleshooting decisions.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led thin-film coating process development, linking CVD/surface-preparation variables to morphology, interface quality, optical performance, and reproducible process controls.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition using DOE strategies and in-situ QCM-D process monitoring to quantify growth kinetics and coating reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Analytical Method Development Bridge BMS LC/LC-MS Match

Connects material-quality questions to LC-MS/GPC/SEC/UV-Vis/FT-IR-style analytical readouts, documentation, troubleshooting, and method-transfer readiness.

LC-MS exposure, GPC/SEC, UV-Vis, FT-IR, SOPs

Analytical Stability and Degradation Study Chemistry of Materials 2025

Mapped degradation and performance failure modes across functional material variants using orthogonal analytical characterization.

UV-Vis-NIR, FT-IR, CV, TGA/DSC, NMR, GPC, MS-informed analysis

Process Development Data Package Current

Built DOE-guided process windows, analytical readouts, and reproducibility criteria for polymer process development and scale-up.

DOE, SPC, FT-IR, thermal analysis, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical Development: LC-MS and GC-MS exposure, GPC/SEC, UV-Vis-NIR, FT-IR, NMR, QCM-D, TGA, DSC, cyclic voltammetry, spectroelectrochemistry; advanced LC/LC-MS method ownership to be bridged
Method Lifecycle & CMC Mindset: Method documentation, reproducibility studies, SOPs, protocols, technical reports, troubleshooting, transfer readiness, data integrity
Materials / Modality Bridge: Organic/materials chemistry, polymers, thin films, stability/degradation analysis, RNA/LNP/antibody analytical platforms to be learned
Data & Communication: DOE, SPC, Excel, Python/basic, JMP/basic, publications, presentations, cross-functional technical summaries

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)